1. Exponential Function and the Chain Rule

The number e is the base of the natural exponential function. The derivative of e^x is itself.

$$\frac{d}{dx}(e^x) = e^x$$
$$\text{If } y = e^{f(x)}, \quad y' = e^{f(x)}\, f'(x)$$

Important: Always multiply by the derivative of the exponent.

$$\frac{d}{dx}(e^{3x}) = e^{3x} \cdot 3 = 3e^{3x}$$
$$\frac{d}{dx}(e^{x^2}) = e^{x^2} \cdot 2x = 2xe^{x^2}$$
$$\frac{d}{dx}(e^{\sin x}) = e^{\sin x} \cos x$$

2. Power Rule and Chain Rule

If $y = [u(x)]^n$, differentiate the outer power first, then the inner expression.

$$\frac{d}{dx}[u(x)]^n = n[u(x)]^{n-1}u'(x)$$
$$\frac{d}{dx}(e^{\sin x} - \cot x)^4 = 4(e^{\sin x} - \cot x)^3 \frac{d}{dx}(e^{\sin x} - \cot x)$$

3. Product Rule and Quotient Rule

For differentiable functions $u(x)$ and $v(x)$:

$$\frac{d}{dx}(uv) = u'v + uv'$$
$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{vu' - uv'}{v^2}, \qquad v \neq 0$$

4. Trigonometric Derivative Rules

$$\frac{d}{dx}\sin x = \cos x \qquad \frac{d}{dx}\cos x = -\sin x$$
$$\frac{d}{dx}\tan x = \sec^2 x \qquad \frac{d}{dx}\cot x = -\csc^2 x$$
$$\frac{d}{dx}\sec x = \sec x \tan x \qquad \frac{d}{dx}\csc x = -\csc x \cot x$$

For any $u(x)$, multiply the trigonometric derivative by $u'(x)$.

Goal: Apply the exponential rule together with power, product, quotient and trigonometric rules.

1. What Is Implicit Differentiation?

An equation is *implicit* when x and y appear together (for example, $x^2 + y^2 = 25$). Differentiate both sides with respect to x and treat y as a function of x.

2. Important Formulas

Expression	Derivative with respect to x
y	y'
y^n	$ny^{n-1}y'$
$\sin y$	$\cos y\, y'$
$\cos y$	$-\sin y\, y'$
$\tan y$	$\sec^2 y\, y'$
$\sec y$	$\sec y \tan y\, y'$
e^y	$e^y y'$
e^{4y}	$4e^{4y}y'$
xy	$y + xy'$
e^{xy}	$e^{xy}(y + xy')$
$\sqrt{x+y}$	$\dfrac{1+y'}{2\sqrt{x+y}}$

3. Solution Procedure

1. Differentiate both sides with respect to x.
2. Insert a factor y' when differentiating an expression involving y.
3. Move every term containing y' to one side.
4. Factor out y' and divide to isolate y'.

Introductory Example: Find y' if $x^2 + y^2 = 25$

$$
\begin{aligned}
x^2 + y^2 &= 25 \\
\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) &= \frac{d}{dx}(25) \\
2x + 2yy' &= 0 \\
2yy' &= -2x \\
y' &= \frac{-2x}{2y}
\end{aligned}
$$

$$\boxed{y' = -\frac{x}{y}}$$

Question 1. $y = e^{3x}$

$$u = 3x, \qquad u' = 3$$
$$y' = e^u \cdot u' \qquad = e^{3x} \cdot 3$$
$$\boxed{y' = 3e^{3x}}$$

Question 2. $y = e^{4x+1}$

$$u = 4x + 1, \quad u' = 4 + 0 = 4$$
$$y' = e^u \cdot u' \qquad = e^{4x+1} \cdot 4$$
$$\boxed{y' = 4e^{4x+1}}$$

Question 3. $y = e^{\sin(3x)}$

$$u = \sin(3x)$$
$$u' = \cos(3x) \cdot \frac{d}{dx}(3x) = 3\cos(3x)$$
$$y' = e^u \cdot u' = e^{\sin(3x)} \cdot 3\cos(3x)$$
$$\boxed{y' = 3e^{\sin(3x)}\cos(3x)}$$

Question 4. $y = e^{\cos x}\tan x$

Use the product rule with $u = e^{\cos x}$ and $v = \tan x$.

$$u' = e^{\cos x}\frac{d}{dx}(\cos x) = -e^{\cos x}\sin x$$
$$v' = \sec^2 x$$
$$y' = u'v + uv'$$
$$= (-e^{\cos x}\sin x)\tan x + e^{\cos x}\sec^2 x$$
$$= e^{\cos x}(\sec^2 x - \sin x \tan x)$$
$$\boxed{y' = e^{\cos x}(\sec^2 x - \sin x \tan x)}$$

Question 5. $y = \dfrac{e^{4x} - \sin x}{\cos x - e^{5x}}$

Use $u = e^{4x} - \sin x$ and $v = \cos x - e^{5x}$.

$$u' = 4e^{4x} - \cos x$$
$$v' = -\sin x - 5e^{5x}$$
$$y' = \frac{vu' - uv'}{v^2}$$
$$= \frac{(\cos x - e^{5x})(4e^{4x} - \cos x)}{(\cos x - e^{5x})^2} - \frac{(e^{4x} - \sin x)(-\sin x - 5e^{5x})}{(\cos x - e^{5x})^2}$$
$$\boxed{y' = \frac{(\cos x - e^{5x})(4e^{4x} - \cos x) + (e^{4x} - \sin x)(\sin x + 5e^{5x})}{(\cos x - e^{5x})^2}}$$

Question 6. $y = e^{4x^2} - 5x + \cos(3x)$

Differentiate each term separately.

$$\begin{aligned}
\frac{d}{dx}(e^{4x^2}) &= e^{4x^2} \cdot \frac{d}{dx}(4x^2) \\
&= e^{4x^2}(8x) \\
\frac{d}{dx}(-5x) &= -5 \\
\frac{d}{dx}(\cos 3x) &= -\sin 3x \cdot \frac{d}{dx}(3x) \\
&= -3\sin 3x \\
y' &= 8xe^{4x^2} - 5 - 3\sin 3x
\end{aligned}$$

$$\boxed{y' = 8xe^{4x^2} - 5 - 3\sin 3x}$$

Question 7. $y = e^{\cot x} - \tan(x^2)$

Apply the chain rule to both terms.

$$\begin{aligned}
\frac{d}{dx}(e^{\cot x}) &= e^{\cot x} \cdot \frac{d}{dx}(\cot x) \\
&= -e^{\cot x}\csc^2 x \\
\frac{d}{dx}(\tan(x^2)) &= \sec^2(x^2) \cdot \frac{d}{dx}(x^2) \\
&= 2x\sec^2(x^2) \\
y' &= -e^{\cot x}\csc^2 x - 2x\sec^2(x^2)
\end{aligned}$$

$$\boxed{y' = -e^{\cot x}\csc^2 x - 2x\sec^2(x^2)}$$

Question 8. $y = (e^{\sin x} - \cot x)^4$

Choose the inner function $u = e^{\sin x} - \cot x$.

$$\begin{aligned}
u' &= \frac{d}{dx}(e^{\sin x}) - \frac{d}{dx}(\cot x) \\
&= e^{\sin x}\cos x - (-\csc^2 x) \\
&= e^{\sin x}\cos x + \csc^2 x \\
y' &= 4u^3 u' \\
&= 4(e^{\sin x} - \cot x)^3(e^{\sin x}\cos x + \csc^2 x)
\end{aligned}$$

$$\boxed{y' = 4(e^{\sin x} - \cot x)^3(e^{\sin x}\cos x + \csc^2 x)}$$

Key Reminder

For $\tan(x^2)$ the argument is x^2: the derivative is $2x\sec^2(x^2)$, not $2x\sec^2 x$.

Question 9. $y = \dfrac{e^{3x^2}\sec x}{x-1}$

Set $u = e^{3x^2}\sec x$ and $v = x-1$. Differentiate the numerator by the product rule.

$$\begin{aligned}
\frac{d}{dx}(e^{3x^2}) &= e^{3x^2}\cdot 6x \\
\frac{d}{dx}(\sec x) &= \sec x\tan x \\
u' &= (6xe^{3x^2})\sec x + e^{3x^2}(\sec x\tan x) \\
&= e^{3x^2}\sec x(6x+\tan x) \\
v' &= 1 \\
y' &= \frac{vu'-uv'}{v^2} \\
&= \frac{(x-1)e^{3x^2}\sec x(6x+\tan x)}{(x-1)^2} \\
&\quad - \frac{e^{3x^2}\sec x\cdot 1}{(x-1)^2} \\
&= \frac{e^{3x^2}\sec x[(x-1)(6x+\tan x)-1]}{(x-1)^2}
\end{aligned}$$

$$\boxed{y' = \frac{e^{3x^2}\sec x[(x-1)(6x+\tan x)-1]}{(x-1)^2}}$$

Question 10. $y = \dfrac{e^{4x^3}(5x^2-1)}{4x^5-4}$

Set $u = e^{4x^3}(5x^2-1)$ and $v = 4x^5-4$.

$$\begin{aligned}
\frac{d}{dx}(e^{4x^3}) &= e^{4x^3}\cdot 12x^2 \\
\frac{d}{dx}(5x^2-1) &= 10x \\
u' &= (12x^2e^{4x^3})(5x^2-1) + e^{4x^3}(10x) \\
&= e^{4x^3}[12x^2(5x^2-1)+10x] \\
v' &= 20x^4 \\
y' &= \frac{vu'-uv'}{v^2} \\
&= \frac{(4x^5-4)e^{4x^3}[12x^2(5x^2-1)+10x]}{(4x^5-4)^2} \\
&\quad - \frac{e^{4x^3}(5x^2-1)(20x^4)}{(4x^5-4)^2}
\end{aligned}$$

$$\boxed{y' = \frac{e^{4x^3}\left[(4x^5-4)(12x^2(5x^2-1)+10x) - 20x^4(5x^2-1)\right]}{(4x^5-4)^2}}$$

Question 11. $x + y = \sin x + \cos y$

$$\begin{aligned}
\frac{d}{dx}(x) + \frac{d}{dx}(y) &= \frac{d}{dx}(\sin x) + \frac{d}{dx}(\cos y) \\
1 + y' &= \cos x - \sin y\, y' \\
y' + \sin y\, y' &= \cos x - 1 \\
y'(1 + \sin y) &= \cos x - 1 \\
y' &= \frac{\cos x - 1}{1 + \sin y}
\end{aligned}$$

$$\boxed{y' = \frac{\cos x - 1}{1 + \sin y}}$$

Question 12. $2x + y^2 = 3x^2 + 4y$

$$\begin{aligned}
\frac{d}{dx}(2x) + \frac{d}{dx}(y^2) &= \frac{d}{dx}(3x^2) + \frac{d}{dx}(4y) \\
2 + 2yy' &= 6x + 4y' \\
2yy' - 4y' &= 6x - 2 \\
y'(2y - 4) &= 6x - 2 \\
y' &= \frac{6x - 2}{2y - 4}
\end{aligned}$$

$$\boxed{y' = \frac{6x - 2}{2y - 4}}$$

Question 13. $e^{xy} + \cos x - \sin y = 0$

Because $(xy)' = y + xy'$, use both product and chain rules.

$$\begin{aligned}
\frac{d}{dx}(e^{xy}) + \frac{d}{dx}(\cos x) - \frac{d}{dx}(\sin y) &= 0 \\
e^{xy}(y + xy') - \sin x - \cos y\, y' &= 0 \\
ye^{xy} + xe^{xy}y' - \sin x - \cos y\, y' &= 0 \\
xe^{xy}y' - \cos y\, y' &= \sin x - ye^{xy} \\
y'(xe^{xy} - \cos y) &= \sin x - ye^{xy} \\
y' &= \frac{\sin x - ye^{xy}}{xe^{xy} - \cos y}
\end{aligned}$$

$$\boxed{y' = \frac{\sin x - ye^{xy}}{xe^{xy} - \cos y}}$$

Question 14. $\sqrt{x+y} = x^2 + y^2$

Write the square root as a power and apply the chain rule.

$$
\begin{aligned}
(x+y)^{1/2} &= x^2 + y^2 \\
\frac{1}{2}(x+y)^{-1/2}(1+y') &= 2x + 2yy' \\
\frac{1+y'}{2\sqrt{x+y}} &= 2x + 2yy' \\
\frac{1}{2\sqrt{x+y}} + \frac{y'}{2\sqrt{x+y}} &= 2x + 2yy' \\
\frac{y'}{2\sqrt{x+y}} - 2yy' &= 2x - \frac{1}{2\sqrt{x+y}} \\
y'\left(\frac{1}{2\sqrt{x+y}} - 2y\right) &= 2x - \frac{1}{2\sqrt{x+y}} \\
y' &= \frac{2x - \frac{1}{2\sqrt{x+y}}}{\frac{1}{2\sqrt{x+y}} - 2y} \\
&= \frac{(2\sqrt{x+y})(2x) - (2\sqrt{x+y})\frac{1}{2\sqrt{x+y}}}{(2\sqrt{x+y})\frac{1}{2\sqrt{x+y}} - (2\sqrt{x+y})(2y)} \\
&= \frac{4x\sqrt{x+y} - 1}{1 - 4y\sqrt{x+y}}
\end{aligned}
$$

$$\boxed{y' = \frac{4x\sqrt{x+y} - 1}{1 - 4y\sqrt{x+y}}}$$

Question 15. $x^3 - 4y^2 - \sin x - e^y = 0$

$$
\begin{aligned}
\frac{d}{dx}(x^3) - 4\frac{d}{dx}(y^2) - \frac{d}{dx}(\sin x) - \frac{d}{dx}(e^y) &= 0 \\
3x^2 - 8yy' - \cos x - e^y y' &= 0 \\
-8yy' - e^y y' &= -3x^2 + \cos x \\
8yy' + e^y y' &= 3x^2 - \cos x \\
y'(8y + e^y) &= 3x^2 - \cos x \\
y' &= \frac{3x^2 - \cos x}{8y + e^y}
\end{aligned}
$$

$$\boxed{y' = \frac{3x^2 - \cos x}{8y + e^y}}$$

Question 16. $\sin x + e^x + \cos y - e^{4y} = 0$

Differentiate each term; include y' for each function of y.

$$\begin{aligned}
\frac{d}{dx}(\sin x) + \frac{d}{dx}(e^x) + \frac{d}{dx}(\cos y) - \frac{d}{dx}(e^{4y}) &= 0 \\
\cos x + e^x - \sin y\, y' - 4e^{4y}y' &= 0 \\
-\sin y\, y' - 4e^{4y}y' &= -\cos x - e^x \\
\sin y\, y' + 4e^{4y}y' &= \cos x + e^x \\
y'(\sin y + 4e^{4y}) &= \cos x + e^x \\
y' &= \frac{\cos x + e^x}{\sin y + 4e^{4y}}
\end{aligned}$$

$$\boxed{y' = \frac{\cos x + e^x}{\sin y + 4e^{4y}}}$$

Question 17. $\tan y + y^4 - 5x = \sec x - y$

$$\begin{aligned}
\frac{d}{dx}(\tan y) + \frac{d}{dx}(y^4) - \frac{d}{dx}(5x) &= \frac{d}{dx}(\sec x) - \frac{d}{dx}(y) \\
\sec^2 y\, y' + 4y^3y' - 5 &= \sec x \tan x - y' \\
\sec^2 y\, y' + 4y^3y' + y' &= \sec x \tan x + 5 \\
y'(\sec^2 y + 4y^3 + 1) &= \sec x \tan x + 5 \\
y' &= \frac{\sec x \tan x + 5}{\sec^2 y + 4y^3 + 1}
\end{aligned}$$

$$\boxed{y' = \frac{\sec x \tan x + 5}{\sec^2 y + 4y^3 + 1}}$$

Common Mistake to Avoid

$\dfrac{d}{dx}e^{4y} = 4e^{4y}y'$ (*not* $4e^{4y}$), because the exponent contains $y(x)$.

Question 18. $y = \left(\dfrac{x^2+\sin x}{x+1}\right)^6$

Use the chain rule, then the quotient rule for the inner function.

$$u = \frac{x^2+\sin x}{x+1}, \qquad y = u^6$$
$$a = x^2+\sin x, \quad a' = 2x+\cos x$$
$$b = x+1, \quad b' = 1$$
$$u' = \frac{ba'-ab'}{b^2}$$
$$= \frac{(x+1)(2x+\cos x)-(x^2+\sin x)(1)}{(x+1)^2}$$
$$y' = 6u^5u'$$
$$= 6\left(\frac{x^2+\sin x}{x+1}\right)^5 \frac{(x+1)(2x+\cos x)-(x^2+\sin x)}{(x+1)^2}$$
$$\boxed{y' = 6\left(\frac{x^2+\sin x}{x+1}\right)^5 \frac{(x+1)(2x+\cos x)-(x^2+\sin x)}{(x+1)^2}}$$

Question 19. $y = (2-5\sin\sqrt{x})^3$

$$u = 2-5\sin\sqrt{x}, \qquad y = u^3$$
$$\frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}}$$
$$u' = -5\cos\sqrt{x}\cdot\frac{1}{2\sqrt{x}}$$
$$= -\frac{5\cos\sqrt{x}}{2\sqrt{x}}$$
$$y' = 3u^2u'$$
$$= 3(2-5\sin\sqrt{x})^2\left(-\frac{5\cos\sqrt{x}}{2\sqrt{x}}\right)$$
$$\boxed{y' = -\frac{15(2-5\sin\sqrt{x})^2\cos\sqrt{x}}{2\sqrt{x}}}$$

Question 20. $y = x\cos^2(3x-1)$

$$u = x, \quad v = [\cos(3x-1)]^2$$
$$u' = 1$$
$$v' = 2\cos(3x-1)[-\sin(3x-1)]\cdot 3$$
$$= -6\cos(3x-1)\sin(3x-1)$$
$$y' = u'v+uv'$$
$$= \cos^2(3x-1) - 6x\cos(3x-1)\sin(3x-1)$$
$$\boxed{y' = \cos^2(3x-1) - 6x\cos(3x-1)\sin(3x-1)}$$

Question 21. $y = x\sec^2(3x-1)$

Use the product and chain rules.

$$\begin{aligned}
u &= x, \quad v = [\sec(3x-1)]^2 \\
u' &= 1 \\
w &= 3x-1, \qquad w' = 3 \\
\frac{d}{dx}[\sec w] &= \sec w \tan w \cdot w' \\
&= 3\sec(3x-1)\tan(3x-1) \\
v' &= 2\sec(3x-1)\frac{d}{dx}[\sec(3x-1)] \\
&= 2\sec(3x-1)[3\sec(3x-1)\tan(3x-1)] \\
&= 6\sec^2(3x-1)\tan(3x-1) \\
y' &= u'v + uv' \\
&= 1\cdot\sec^2(3x-1) + x\cdot 6\sec^2(3x-1)\tan(3x-1)
\end{aligned}$$

$$\boxed{y' = \sec^2(3x-1) + 6x\sec^2(3x-1)\tan(3x-1)}$$

Question 22. $y = \csc^3(4x^2-1)$

Apply the chain rule to the outer power, trigonometric function and inner expression.

$$\begin{aligned}
u &= 4x^2-1 \\
u' &= 8x \\
v &= \csc u, \qquad y = v^3 \\
\frac{dv}{dx} &= -\csc u\cot u\cdot u' \\
&= -8x\csc(4x^2-1)\cot(4x^2-1) \\
y' &= 3v^2v' \\
&= 3\csc^2(4x^2-1) \\
&\quad\cdot[-8x\csc(4x^2-1)\cot(4x^2-1)] \\
&= -24x\csc^3(4x^2-1)\cot(4x^2-1)
\end{aligned}$$

$$\boxed{y' = -24x\csc^3(4x^2-1)\cot(4x^2-1)}$$

Review Checklist

- Chain rule: multiply by the derivative of the inside expression.
- Product rule: differentiate each factor once, then add.
- Quotient rule: denominator times numerator derivative minus numerator times denominator derivative.
- Implicit rule: include y' every time a function of y is differentiated.

1. Problem

Find $\dfrac{dy}{dx}$ by implicit differentiation.

$$\tan y + y^4 - 5x = \sec x - y$$

2. Differentiate Both Sides

Differentiate both sides with respect to x.

$$\sec^2(y)\, y' + 4y^3 y' - 5 = \sec x \tan x - y'$$

3. Move the y' Term to the Left

Add y' to both sides to move the y' term from the right to the left.

$$\sec^2(y)\, y' + 4y^3 y' + y' = \sec x \tan x + 5$$

4. Factor Out y'

Factor out y' on the left side.

$$y'\left(\sec^2(y) + 4y^3 + 1\right) = \sec x \tan x + 5$$

5. Solve for y'

Divide both sides by $\sec^2(y) + 4y^3 + 1$.

$$\boxed{y' = \frac{\sec x \tan x + 5}{\sec^2 y + 4y^3 + 1}}$$

Key Idea

Every derivative involving y must be multiplied by y' when using implicit differentiation.

Example 1. $x^3 - 4y^2 - \sin x - e^y = 0$

Differentiate both sides of the equation with respect to x.

$$3x^2 - 8y\,y' - \cos x - e^y y' = 0 \qquad \text{(differentiate each term)}$$

$$3x^2 - \cos x - 8y\,y' - e^y y' = 0 \qquad \text{(collect } y' \text{ terms)}$$

$$(-8y - e^y)\,y' + 3x^2 - \cos x = 0 \qquad \text{(factor } y')$$

$$(-8y - e^y)\,y' = -3x^2 + \cos x \qquad \text{(solve for } y')$$

$$\boxed{y' = \frac{3x^2 - \cos x}{8y + e^y}}$$

Example 2. $\sin x + e^x + \cos y - e^{4y} = 0$

Differentiate both sides of the equation with respect to x.

$$\cos x + e^x - \sin y\,y' - 4e^{4y} y' = 0 \qquad \text{(differentiate each term)}$$

$$\cos x + e^x - (\sin y + 4e^{4y})\,y' = 0 \qquad \text{(collect } y' \text{ terms)}$$

$$-(\sin y + 4e^{4y})\,y' + \cos x + e^x = 0 \qquad \text{(factor } y')$$

$$-(\sin y + 4e^{4y})\,y' = -(\cos x + e^x) \qquad \text{(solve for } y')$$

$$\boxed{y' = \frac{\cos x + e^x}{\sin y + 4e^{4y}}}$$

11 Implicit Differentiation – Detailed Example

Implicit Differentiation

Example 1. Find y' if $\sqrt{x+y} = x^2 + y^2$.

Find y' if $\sqrt{x+y} = x^2 + y^2$.

1. Differentiate both sides with respect to x.

Differentiate both sides of $\sqrt{x+y} = x^2 + y^2$ with respect to x, using the chain rule.

$$\frac{d}{dx}\left(\sqrt{x+y}\right) = \frac{d}{dx}(x^2 + y^2).$$

2. Apply the derivative formulas.

Using the chain rule, the derivative of $\sqrt{x+y}$ is $\dfrac{1}{2\sqrt{x+y}}(1+y')$, and the right side differentiates to $2x + 2yy'$.

$$\frac{1+y'}{2\sqrt{x+y}} = 2x + 2yy'.$$

3. Multiply out or rearrange carefully.

Multiply both sides by $2\sqrt{x+y}$ to clear the denominator.

$$1 + y' = (2x + 2yy')(2\sqrt{x+y}).$$

$$1 + y' = 4x\sqrt{x+y} + 4yy'\sqrt{x+y}.$$

4. Collect the terms containing y' on one side.

Move the terms with y' to the left side and the constant term to the right side.

$$y' - 4yy'\sqrt{x+y} = 4x\sqrt{x+y} - 1.$$

5. Factor out y'.

Factor y' from the left side.

$$y'\left(1 - 4y\sqrt{x+y}\right) = 4x\sqrt{x+y} - 1.$$

6. Solve for y'.

Divide both sides by $1 - 4y\sqrt{x+y}$.

$$y' = \frac{4x\sqrt{x+y} - 1}{1 - 4y\sqrt{x+y}}.$$

7. Final answer (two equivalent forms).

We can also express the answer in the form obtained before clearing the denominator.

$$y' = \frac{2x - \dfrac{1}{2\sqrt{x+y}}}{\dfrac{1}{2\sqrt{x+y}} - 2y} = \boxed{y' = \frac{4x\sqrt{x+y} - 1}{1 - 4y\sqrt{x+y}}}$$

14 Mixed Differentiation Examples

Implicit Differentiation

Example 1. $y = \left(\dfrac{x^2+\sin x}{x+1}\right)^6$

Let $u = \dfrac{x^2+\sin x}{x+1}$. Then $y = u^6$.

Differentiate using the chain rule: $\dfrac{dy}{dx} = 6u^5\dfrac{du}{dx}$.

Find u' using the quotient rule. Let $f(x) = x^2+\sin x$ and $g(x) = x+1$. Then $f'(x) = 2x+\cos x$ and $g'(x) = 1$.

$$\begin{aligned} u' &= \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2} \\ &= \frac{(x+1)(2x+\cos x) - (x^2+\sin x)(1)}{(x+1)^2} \\ &= \frac{(x+1)(2x+\cos x) - (x^2+\sin x)}{(x+1)^2} \end{aligned}$$

Substitute u and u' into $y' = 6u^5u'$:

$$y' = 6\left(\frac{x^2+\sin x}{x+1}\right)^5\left(\frac{(x+1)(2x+\cos x) - (x^2+\sin x)}{(x+1)^2}\right).$$

$$\boxed{y' = 6\left(\frac{x^2+\sin x}{x+1}\right)^5\frac{(x+1)(2x+\cos x) - (x^2+\sin x)}{(x+1)^2}}$$

Example 2. $y = \left(2-5\sin(\sqrt{x})\right)^3$

Let $u = 2-5\sin(\sqrt{x})$. Then $y = u^3$.

Differentiate using the chain rule: $\dfrac{dy}{dx} = 3u^2\dfrac{du}{dx}$.

Find u'. $\dfrac{du}{dx} = -5\dfrac{d}{dx}[\sin(\sqrt{x})]$.

Use the chain rule for $\sin(\sqrt{x})$: $\dfrac{d}{dx}[\sin(\sqrt{x})] = \cos(\sqrt{x})\cdot\dfrac{d}{dx}[\sqrt{x}]$.

The derivative of $\sqrt{x} = x^{1/2}$ is $\dfrac{d}{dx}[\sqrt{x}] = \dfrac{1}{2}x^{-1/2} = \dfrac{1}{2\sqrt{x}}$.

Therefore, $\dfrac{du}{dx} = -5\cos(\sqrt{x})\cdot\dfrac{1}{2\sqrt{x}} = \dfrac{-5\cos(\sqrt{x})}{2\sqrt{x}}$.

Substitute u and u' into $y' = 3u^2u'$:

$$y' = 3\left(2-5\sin(\sqrt{x})\right)^2\left(-\frac{5\cos(\sqrt{x})}{2\sqrt{x}}\right)$$

$$\boxed{y' = -\frac{15\left(2-5\sin(\sqrt{x})\right)^2\cos(\sqrt{x})}{2\sqrt{x}}}$$

15 Final Worked Examples and Review

Implicit Differentiation

Example 1. $y = x\cos^2(3x-1)$

Use the product rule and chain rule.

$$y = x\cos^2(3x-1)$$

$$y' = 1\cdot\cos^2(3x-1) + x\cdot\frac{d}{dx}\left[\cos^2(3x-1)\right]$$

$$= \cos^2(3x-1) + x\cdot 2\cos(3x-1)\cdot(-\sin(3x-1))\cdot 3 \qquad \text{(chain rule)}$$

$$= \cos^2(3x-1) - 6x\cos(3x-1)\sin(3x-1)$$

$$\boxed{y' = \cos^2(3x-1) - 6x\cos(3x-1)\sin(3x-1)}$$

Example 2. $y = x\sec^2(3x-1)$

Use the product rule and chain rule.

$$y = x\sec^2(3x-1)$$

$$y' = 1\cdot\sec^2(3x-1) + x\cdot\frac{d}{dx}\left[\sec^2(3x-1)\right]$$

$$= \sec^2(3x-1) + x\cdot 2\sec^2(3x-1)\tan(3x-1)\cdot 3 \qquad \text{(chain rule)}$$

$$= \sec^2(3x-1) + 6x\sec^2(3x-1)\tan(3x-1)$$

$$\boxed{y' = \sec^2(3x-1) + 6x\sec^2(3x-1)\tan(3x-1)}$$

Example 3. $y = \csc^3(4x^2-1)$

Use the chain rule.

$$y = \csc^3(4x^2-1)$$

$$y' = 3\csc^2(4x^2-1)\cdot\frac{d}{dx}\left[\csc(4x^2-1)\right]$$

$$= 3\csc^2(4x^2-1)\cdot\left(-\csc(4x^2-1)\cot(4x^2-1)\right)\cdot(8x)$$

$$= -24x\csc^3(4x^2-1)\cot(4x^2-1)$$

$$\boxed{y' = -24x\csc^3(4x^2-1)\cot(4x^2-1)}$$

End of Lecture Summary

1. Differentiate $e^{f(x)}$ as $e^{f(x)}f'(x)$.
2. In implicit differentiation, every derivative of y includes y'.
3. Collect all terms with y' on one side of the equation.
4. Factor and solve for y'.